Virtual Lab Cybersecurity Handbook

Using VMware Workstation Pro, Kali Linux, and Windows 10.

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Course: ENGL&235 – Technical Writing

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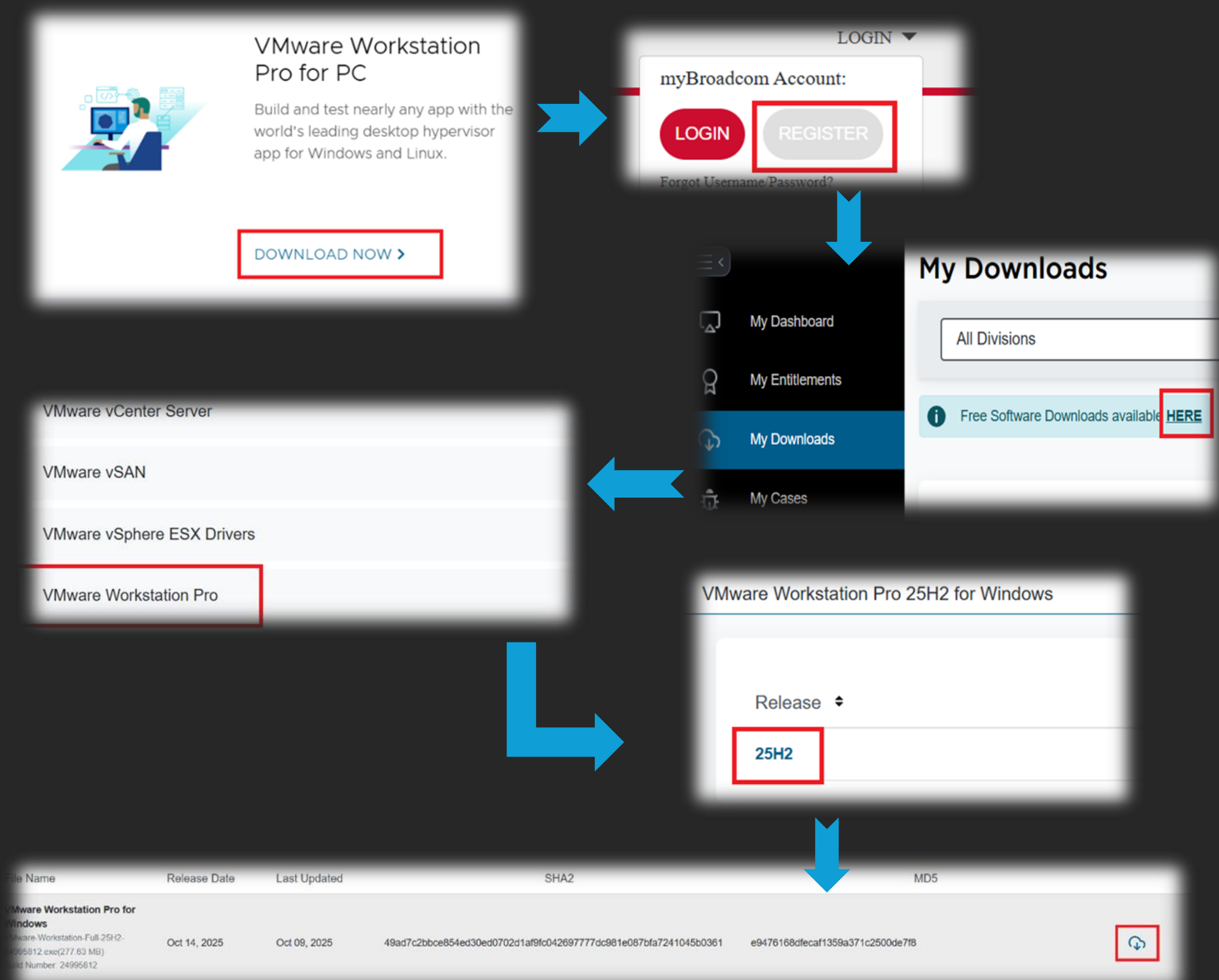
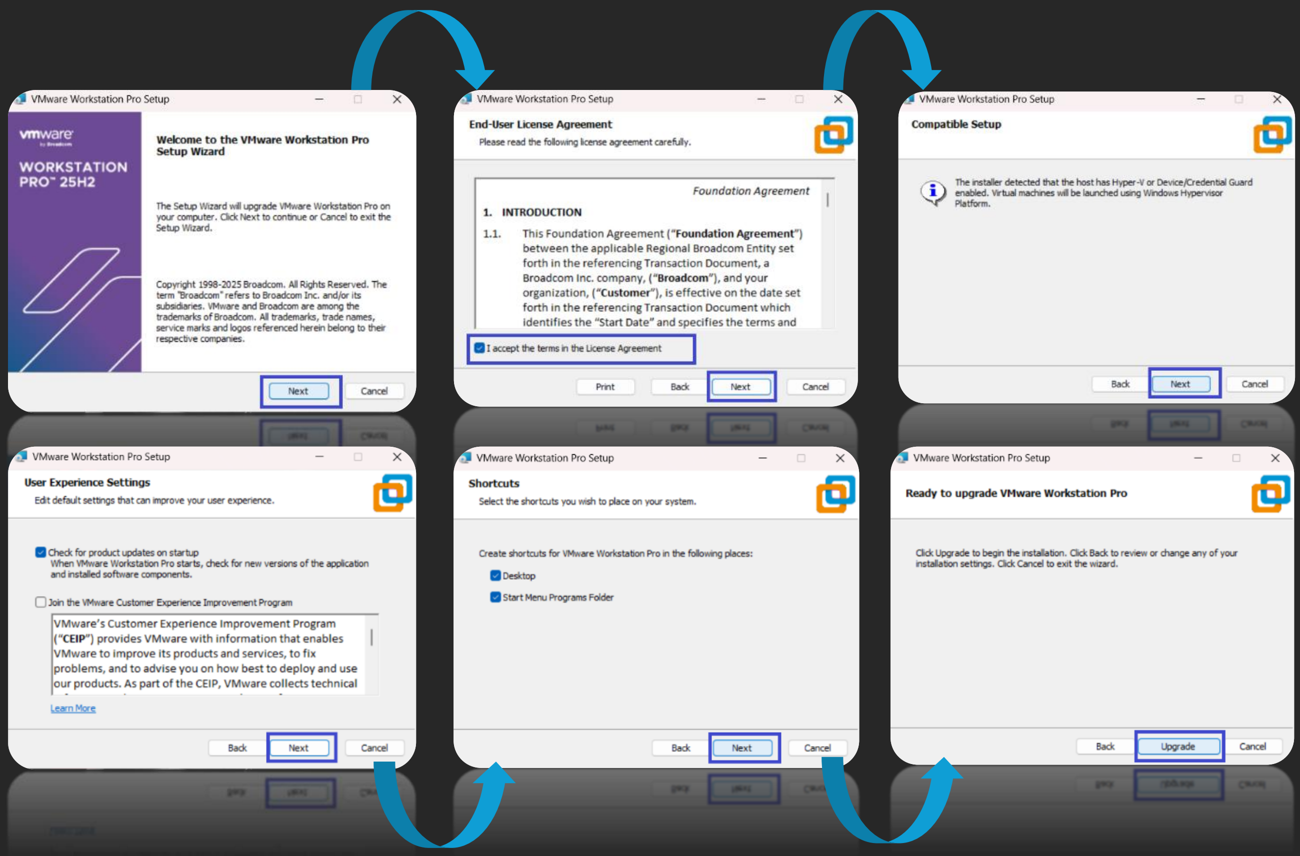
Install VMware on your windows 11 machine:

The first step of this process is to install your windows hypervisor to manage your virtual machines. There are many options to choose from, Hyper V, VMware, and Virtualbox are among the most popular. In this instance we are going to be working with VMware. In the first step you will install the software and run the setup wizard.

Step 1:

- First, you will need to navigate to https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion
- There you will see a link to download Vmware workstation pro, click download now.
- After clicking download now you will be redirected to a Broadcom webpage where you will need to register for an account.
- Once registered, navigate to my downloads tab on the left-hand side.
- Here you will see a blue pop-up box that says “Free Software Downloads available here” click the word here.
- You will need to accept the terms and conditions in the top left corner then click the download button on the right-hand side.
- After downloading, navigate to your downloads folder and run the installer.





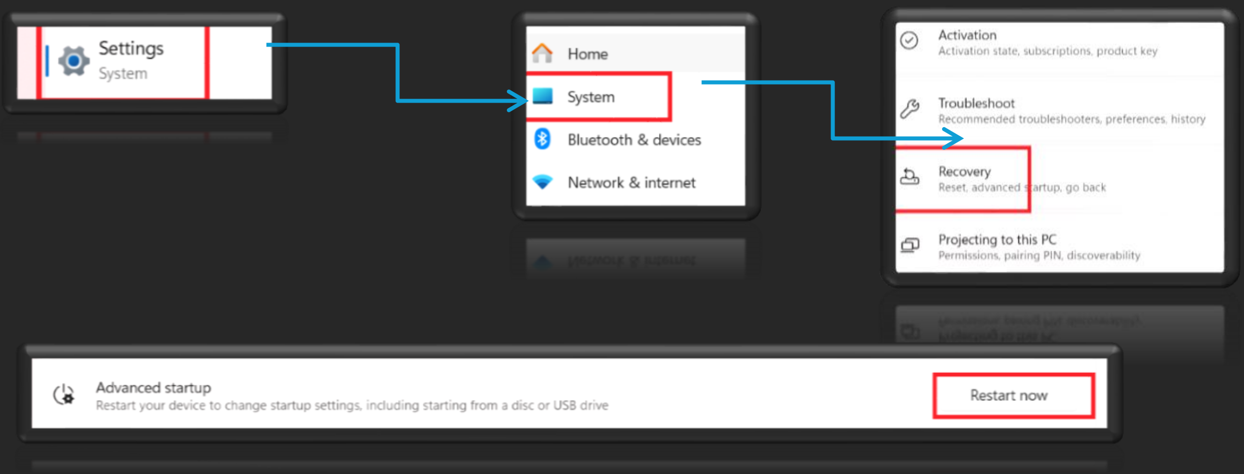
Enable virtualization in your bios:

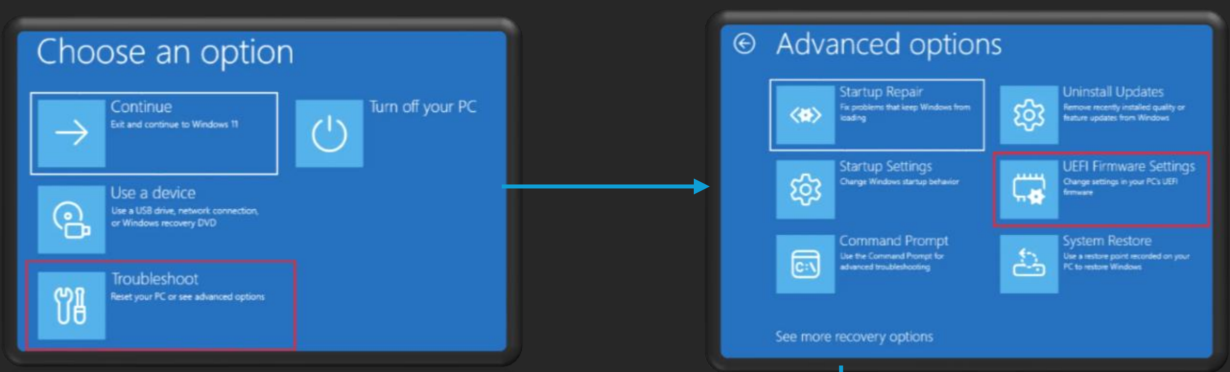
To run virtual machines on windows you must enable virtualization, a setting that can only be enabled from your BIOS menu. You can quickly check if this feature is enabled by going to your task manager and navigating to the performance tab. In the following step I will walk you through a quick way to access your BIOS menu but, keep in mind that the interface of your menu may look different depending on the make and model of your motherboard. If you can't find the setting, you may need to do some research on your specific motherboard.

Step 2:

- Navigate your settings in your windows machine.
- Click system on the left-hand side.
- Scroll down and click on recovery.
- You will see Advanced startup, to the right of that will be a restart now button, click it. This will restart your machine into safe mode, and you will be able to access your bios settings.
- When your machine starts up you will see a blue screen with a few options, look for troubleshoot and click on it.

- Now click on UEFI Firmware settings, this will launch your bios.
- Navigate to your advanced settings and look for SVM and enable this setting.
- Look for save and exit, you will now need to restart your machine and boot up normally.





Note: BIOS interface will look different depending on mother board manufacture, be sure to research where your virtualization settings are in your motherboard bios.

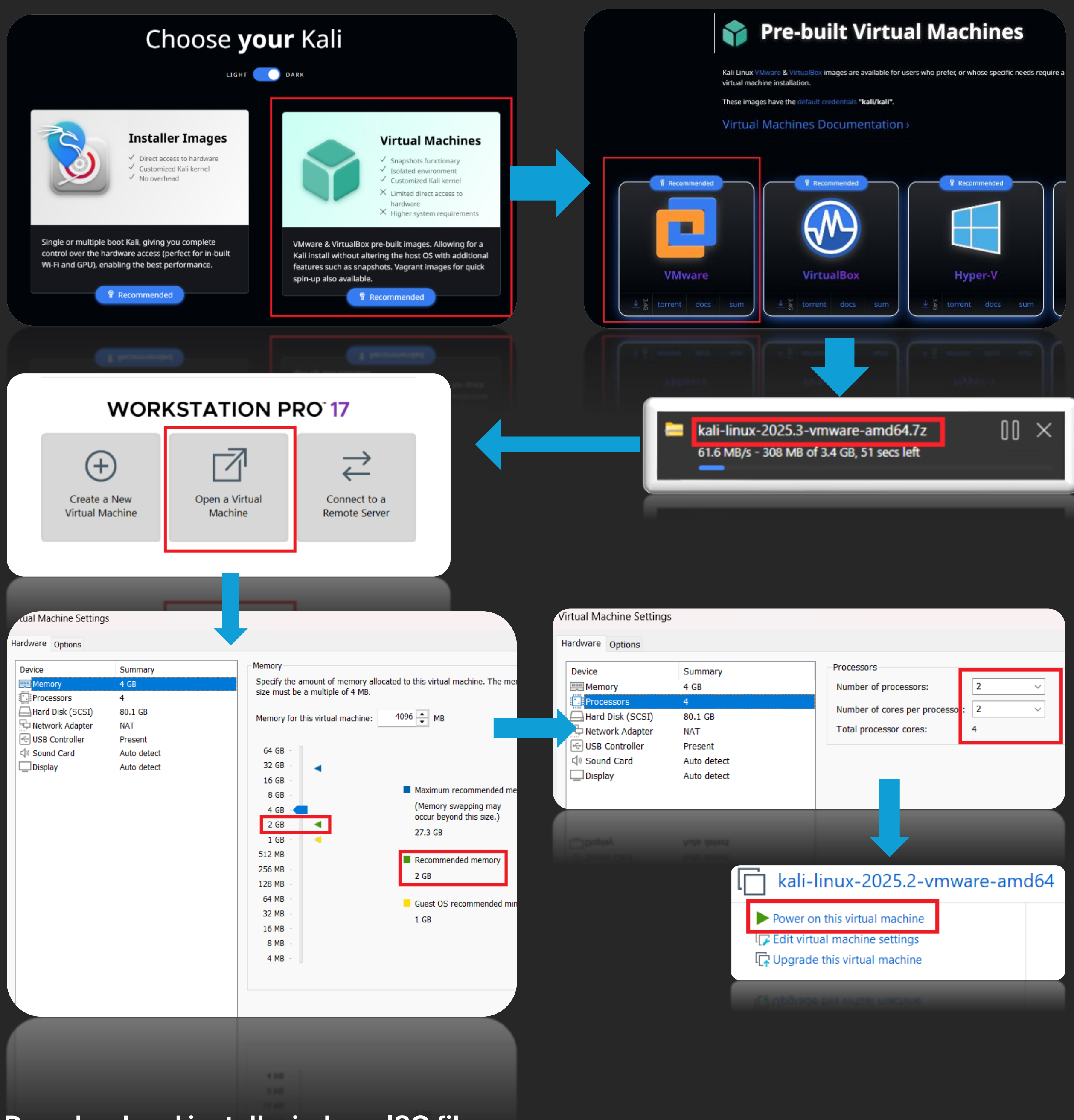
Download and install Kali Linux VM:

In this step you will now download a Kali Linux ISO file. This is a file that will be installed into your hypervisor, it contains a pre-configured kali Linux virtual machine. You will have to navigate to the Kali Linux official website and download the zip folder, extract the contents, open the .vmx file in your hypervisor, and configure how much RAM and storage space you want to allocate to the machine. Depending on how powerful your host machine is, settings may vary. I recommend allocating 2-4 GB of RAM, 4 CPU processors, and 60-80 GB of storage. Remember, a virtual machine pulls resources from your host machine and uses them to run the virtual machine. Make sure you open the file with the vmx extension.

Step 3:

- Navigate to https://www.kali.org/get-kali/#kali-platforms to download the ISO file for Kali Linux.
- You will see two options, Installer Images and Virtual Machines, Select Virtual Machines.
- Now you will see different hypervisors, select VMware.
- This will download a zip file which you will need to extract the contents, do so now.
- Open Vmware Workstation Pro.
- Select Open Virtual Machine.
- Navigate to the directory that you extracted the contents of the zip file too, likely your downloads folder.
- Locate the kali-linux-2025.2-vmware-amd64.vmx file select it, and open.
- On the left-hand side you will see your virtual machine, select it.
- Now look toward the center of your screen and you will see three options, power on, edit virtual machine settings, and upgrade, select edit virtual machine settings.
- Change the memory to 2GB.
- Change Processors to 4
- Change hard disk to 80GB
- Exit virtual machine settings, select power on this virtual machine.
- Your VM will boot up for the first time and bring you to a log in screen, the default username and password are both “Kali”
- You have now successfully installed your first VM

Images on next page...



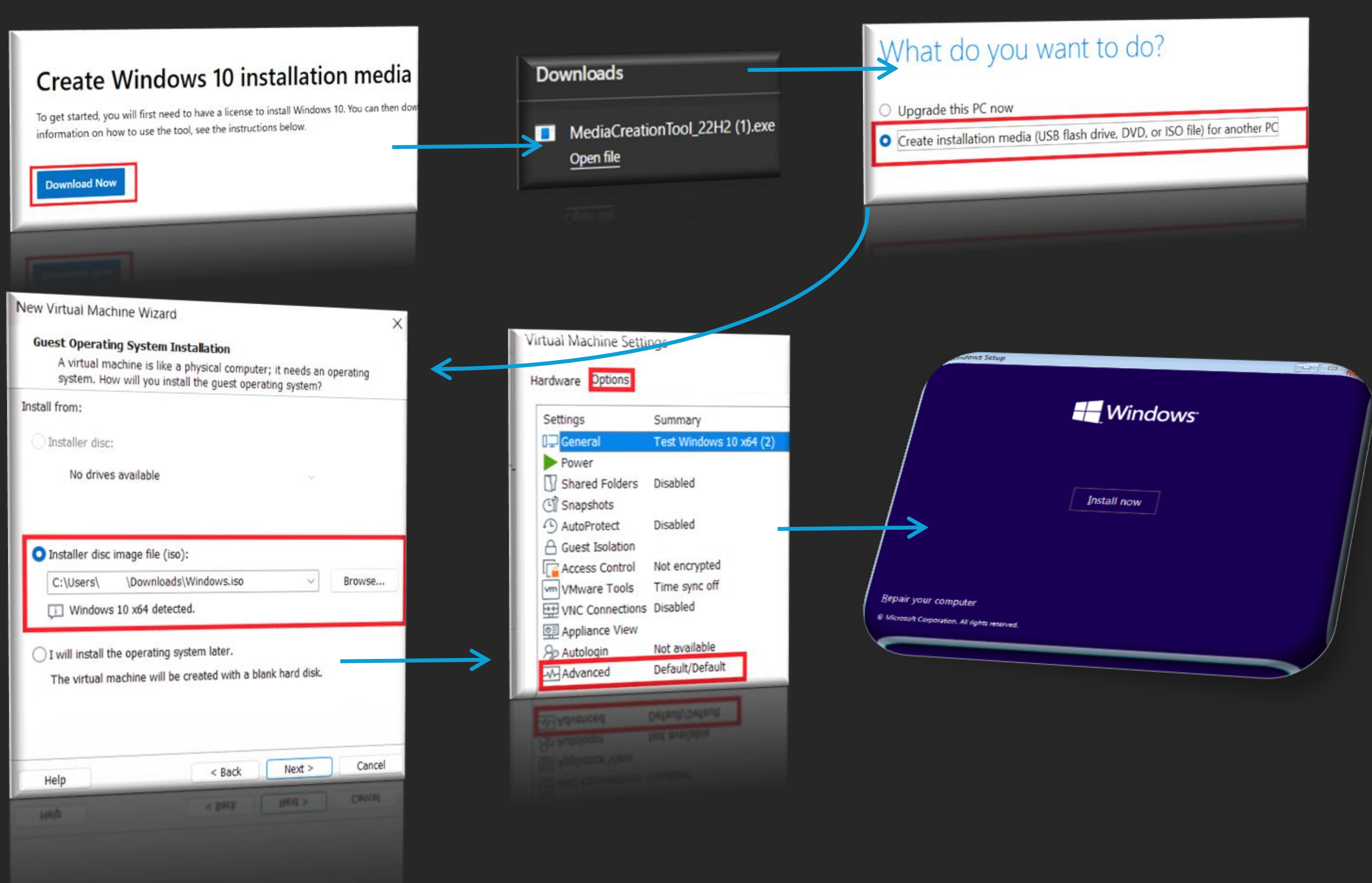
Download and install windows ISO file:

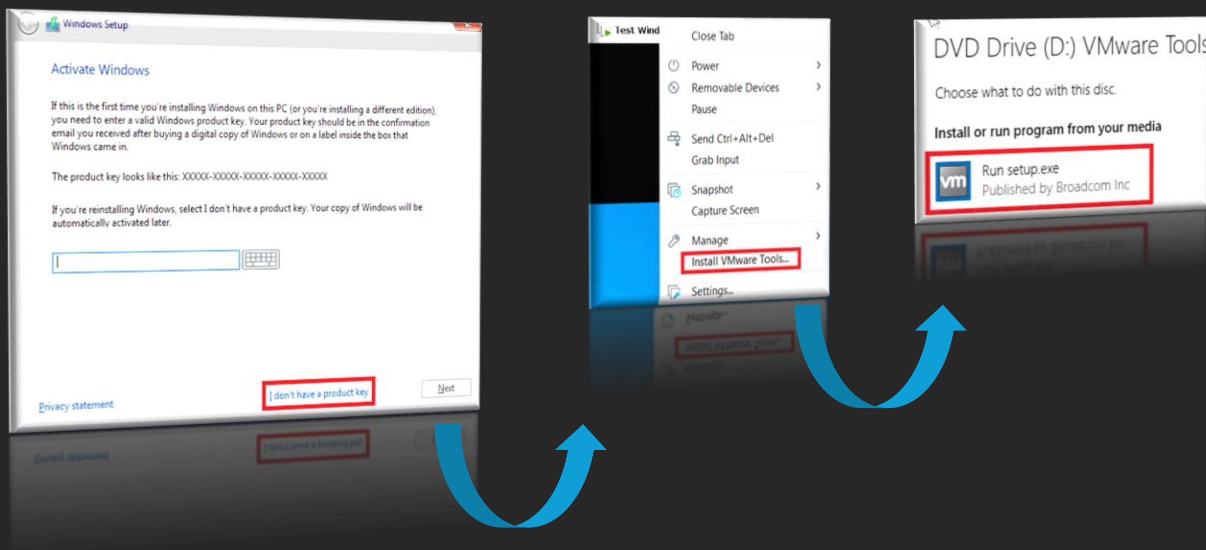
Just like the previous Kali Linux machine, you will need to go to the Microsoft windows official website and download the windows ISO file. Download the create windows 10 installation media and run the setup wizard. Open the created windows ISO file in your hypervisor and allocate resources like before. You will run into an EFI error code when you first boot up the machine, resolve this by powering off the machine and changing the firmware type from UEIF to BIOS. Unlike the Linux machine you will have to install the windows operating system onto this VM, I include steps in step 4.

Step 4: Download and install windows ISO file

- Navigate to https://www.microsoft.com/en-us/software-download/windows10 to download the ISO file for windows 10.
- Look for "Create Windows 10 installation media" and click download now.
- Open the Mediacreationtool_22H2.exe.
- Now accept the terms and conditions.
- Select create installation media and click next.
- Uncheck the box that says use recommended options. Select your preferred language, windows 10 and 64-bit then click next.
- Select ISO file and click next.
- Choose the file path you want to save the ISO file, keeping it in your downloads folder is also fine. Click save.
- Click Finish.
- Open VMware Workstation Pro.
- Select Create New Virtual Machine.
- Run this installer, select typical and click next.
- Select Install Disk Image file, browse to the directory where your ISO file is located then select open and click next.
- Name your machine and click next.
- Select your disk space, 60GB is recommended, choose split virtual disk into multiple files and click next.
- Customize your hardware, I recommend 4GB of ram and 2 Processors.
- Upon boot up of the machine you see a black screen that says, "Press any key to boot from CD or DVD", this will take a couple of minutes then you will receive a EFI network error, just power off the virtual machine.
- While the machine is powered off select edit virtual machine settings.
- At the top of the window select the options tab.
- Look toward the bottom and double click on advanced.
- Now, at the advanced tab on the right-hand side you will see Firmware type change this from UEIF to BIOS.
- Exit and power back on the machine.
- You will now be greeted with the windows installation menu select your language and continue.
- Click install now.
- If you have a product key enter it now if not select, I don't have a product key at the bottom.

- Select the version of windows that you prefer. I'm going with windows 10 pro.
- Installation will take a few minutes.
- Select your region and click continue.
- Choose to set up for personal use.
- Given that this is a virtual machine you can select Offline Account in the bottom left.
- Enter any name you like, this will be the username you login to your VM with.
- Create a password.
- Choose your security questions.
- In the browser settings select not now.
- In the privacy settings turn all these settings off and select accept.
- Once logged into your VM you will be at the desktop but now move your mouse cursor around to your hypervisor (VMware) right click the tab at the top that says the name of your VM.
- Select install VMware tools
- Inside your VM now select Run setup.exe
- Select yes at the first pop up window.
- You can keep everything default on this installation wizard and continue to click next until the installation process has completed.
- Select finish.
- You are now prompted that your VM needs to restart, click yes.





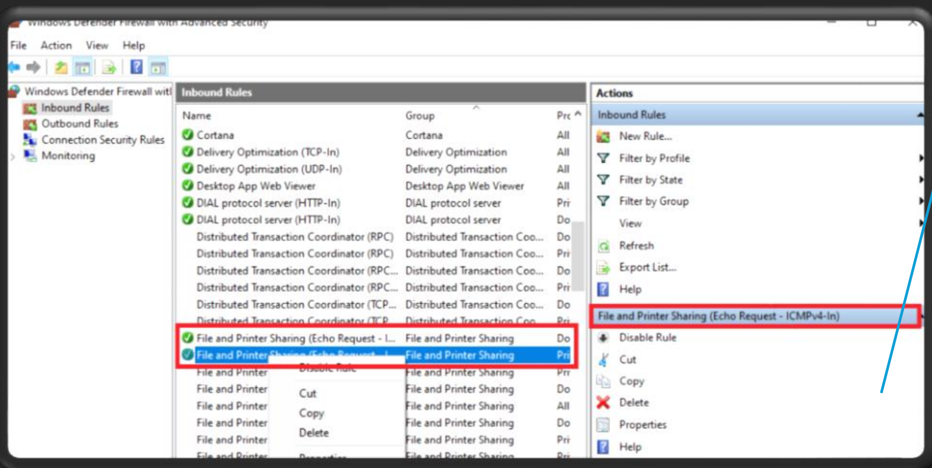
Ensure connectivity between the machines:

For you to use this setup as a lab you need to ensure that the two virtual machines you installed can communicate with each other and are on the same network. You will run a series of ping commands through the Linux terminal and windows command prompt. By default, windows blocks ping requests so you will need to set a rule in your firewall to allow these requests.

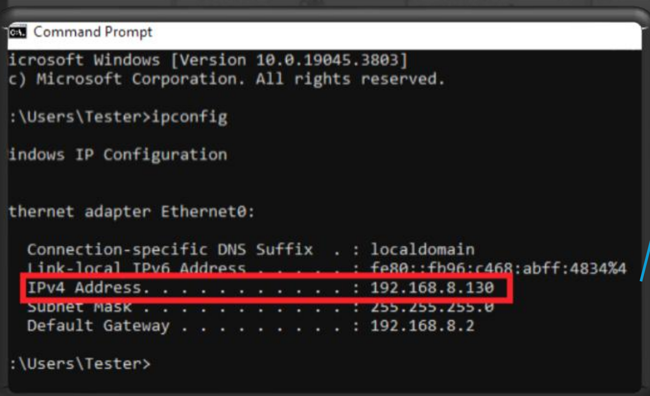
Step 5: Ensure connectivity between the machines

- Launch Vmware Workstation Pro.
- Power on your Linux VM and Power on your Windows VM
- On your windows machine search Windows Firewall Defender Firewall Advanced Security.
- On the left-hand side click Inbound Rules.
- Scroll down until you see File and printer sharing (Echo requests - ICMPv4-in)
- There will be two rules, right click each rule and select enable.
- In your windows machine go to search at the bottom type in cmd, open your command prompt.
- Run command “ipconfig” press enter.
- Take note of the IPv4 Address.
- On your Kali Linux machine open your terminal
- Run command ping -c 4 <Windows IP address> (Note: Make sure you replace windows ip address with the ipv4 address you took note of in the previous step.)
- You should see 4 packets transmitted, 4 packets received. This ensures that the kali linux machine can communicate with the windows machine.
- On the kali linux machine run command “ip a”

- By eth0 you will see inet followed by an IPv4 address and a subnet. Take note of this IP address.
- On your windows machine open your command prompt.
- Run command ping <Kali linux IPv4 address>
- Again, you will see 4 packets sent, 4 packets received.

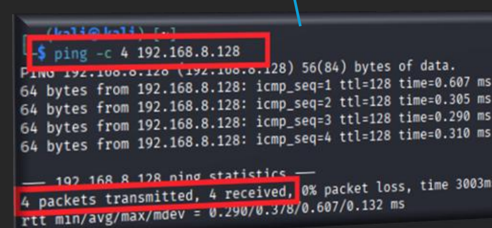


Note: Make sure these rules are enabled, windows defender blocks ping requests by default.

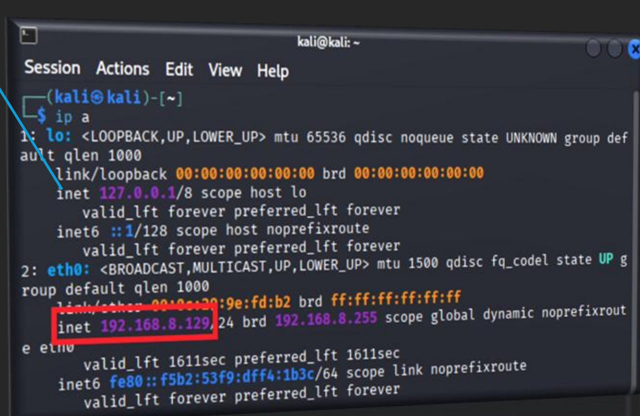


Note: Write this IP address down in a notepad so you can ping it from Linux machine.

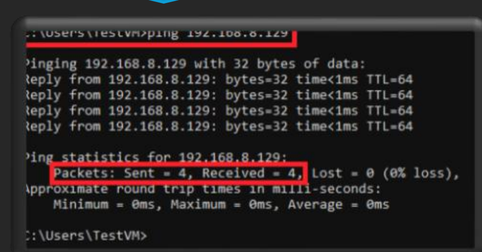
This shows successful ping requests.



Take note of this IP to run ping command on from windows



Note: Your IP Address will be different than the screenshots!



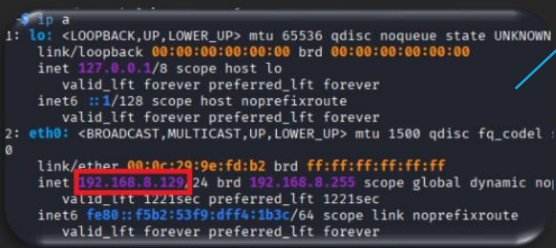
Demonstrating how to simulate an attack using your virtual machines:

Now that you have the two machines set up and ready to run, let's showcase some of the things that this can be used for. The Linux ISO file that you installed comes equipped with tons of reconnaissance and hacking tools that you can use to simulate attacks on the windows machine. You can adjust configurations, turn firewalls on and off, adjust firewall inbound and outbound rules. We are going to be looking for a tool called Nmap, a reconnaissance tool used to scan open ports, running services, and locates IP address. After that we will perform a simple LLMNR Poisoning attack which we obtain a NTLMv2 hash, and a username. We crack the password using a tool called hashcat obtaining the login credentials for the user on that windows machine.

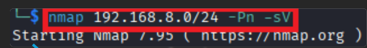
Step 6:

- Power on your Linux and Windows Virtual Machines.
- Open your terminal in Linux.
- Run command "ip a"
- You will see your IPv4 address under eth0 it will look like this "192.168.7.122/24"
- Run Nmap scan to scan your whole subnet for active devices (your looking for your windows VM) Run command nmap <Linux IP/subnet> -Pn -sV example: nmap 192.168.7.0/24 -Pn -sV
- Switch to your windows VM and open your command prompt.
- Run command ipconfig.
- Locate your IPv4 Address.
- Switch back to your linux machine, you will see your windows machine IPv4 in your nmap scan results.
- On your linux terminal type cd /usr/share/responder.
- Run command "sudo python3 Responder.py -I eth0"
- Switch over to your windows machine and press windows key + R
- Type //fakeserver, press enter.
- Switch back to your linux machine and look at the output in your terminal.
- Scroll down until you see "[SMB] NTLMv2-SSP Hash: "
- Highlight the whole hash and copy it.
- Press CTRL + C to cancel the listener.
- Back at the command line, type nano hash.txt to create a new txt file.
- Paste the hash.
- Press CTRL + X on your keyboard.
- Press Y on your keyboard.

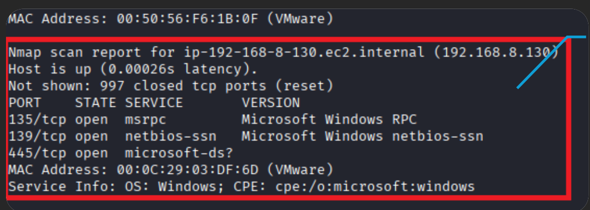
- Press Enter on your keyboard.
- Run this command “hashcat -m 5600 -a 0 hash.txt /usr/share/wordlists/rockyou.txt”
- This should give you an output of the hash that you copied followed by: [your password here].



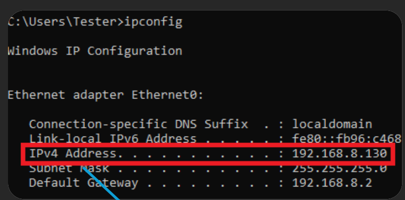
This is step 1, locate the IP address and subnet of your linux machine.



Step 2, run this command replacing the last octet of your IP address with 0 followed by / and your subnet.



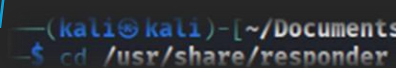
You will see an output like this and locate the one running windows.

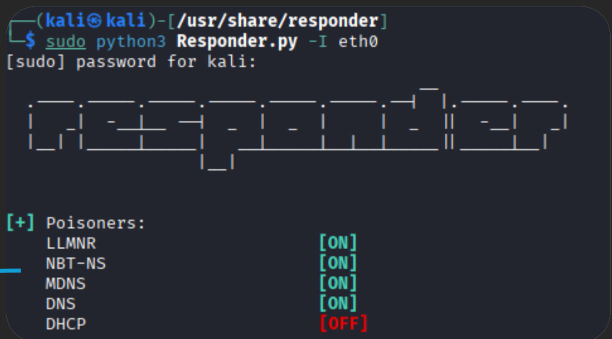


Running ipconfig on windows VM can confirm that is the machine in the scan.

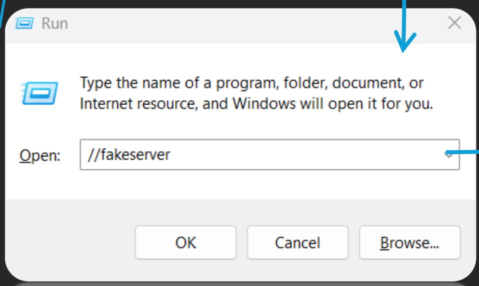
LLMNR Poisoning Attack

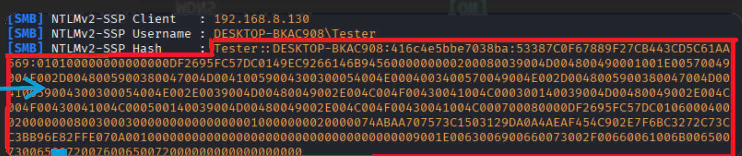
Change to responder directory!





Run this on windows VM





Copy the whole NTLMv2-SSP hash. This includes all the orange text after the : now paste into hash.txt file